Technical Reflection

Mathematical and Analytical Skills Demonstrated

Throughout this project, I demonstrated my understanding of calculus and numerical analysis, particularly the relationship between the difference quotient and the definition of differentiability. The application uses left- and right-hand derivative approximations to investigate whether a function is likely differentiable at a selected point. I also applied analytical reasoning when evaluating the accuracy of these approximations and determining how numerical restrictions influence the application's conclusions.

Why I Selected This Project

I selected Project 3 because it provided more opportunities for expansion, particularly in mathematical exploration and programming. The Differentiability Application offered a foundation that I could continue developing by improving the numerical calculations, expanding the mathematical explanations, and refining the overall functionality of the program.

How I Improved and Expanded the Project

I revised the application's HTML and JavaScript code to establish more restrictive numerical criteria for identifying differentiability. These changes included allowing smaller step sizes, with a minimum value of $h=0.001$, and reducing the tolerance used to compare the left- and right-hand difference quotients.

I also revised the application's Theory page by providing clearer explanations of the difference quotient and the mathematical definition of differentiability. These improvements were influenced by feedback received on the original Project 3 assignment.

Skills Gained from Revising the Project

Revisiting this project helped strengthen my understanding of the difference quotient and its connections to other concepts in numerical analysis, including Taylor series and approximation errors. It also encouraged me to consider the limitations of numerical calculations, particularly how step size and tolerance influence the reliability of results.

Additionally, the revision process helped me improve my HTML and JavaScript skills, gain experience debugging code, and better understand how mathematical concepts can be translated into functional computer programs.

How This Portfolio Supports My Professional Goals

I plan to continue developing my career in mathematics and data analytics. This repository serves as an ongoing portfolio of my mathematical studies, programming experience, and computational projects.

By maintaining and expanding this portfolio, I can demonstrate my technical abilities and academic development to potential employers and other individuals interested in my work. It also provides an opportunity to document my progress as I continue learning new mathematical and programming techniques.